Lesson 7.7 Guided Notes

Session 1 · Polygons and Perimeter on the Plane

I can find the perimeter and area of a polygon drawn on the coordinate plane. · 6.G.A.3

Word / Spanish	Meaning — complete the blank	Example
polygon / polígono	A closed shape with _____ sides.	triangle, rectangle, square
vertex / vértice	A _____ point of a polygon.	(−2, 3)
perimeter / perímetro	The distance all the way _____ a shape.	add every side length
area / área	The number of _____ units inside a shape.	rectangle: length × width

Key idea

Two vertices that share a coordinate make a _____ or vertical side.

Perimeter is the _____ of all the sides. Area of a rectangle is length _____ width.

Worked example

A rectangle has vertices (−1, 4), (5, 4), (5, −3) and (−1, −3). Find its perimeter and area.

Top side: same y, opposite signs, so 1 + 5 = _____ units.

Right side: same x, opposite signs, so 4 + 3 = _____ units.

Perimeter: 6 + 7 + 6 + 7 = _____ units.

Answer: Area = 6 × 7 = _____ square units

You try

1. A square has vertices (−3, 2), (1, 2), (1, −2) and (−3, −2). Find its perimeter.

_____

2. Three corners of a rectangle are (−4, 1), (3, 1) and (3, −5). Name the fourth corner.

_____

Sum it up: To find the perimeter of a polygon on the coordinate plane, I ___ because ___.

_____

Session 2 · Designing Polygons with Scale

I can use a scale to find a missing dimension of a figure on the coordinate plane. · 6.G.A.3

Word / Spanish	Meaning — complete the blank	Example
scale / escala	How much real length one grid _____ stands for.	1 unit = 4 meters
area / área	The flat space _____ a shape, in square units.	20 m × 12 m = 240 m²
dimensions / dimensiones	The _____ and width of a shape.	24 ft by 16 ft

Key idea

Multiply grid units by the _____ to get real length.

To find a missing side, _____ the area by the known side.

Worked example

A rectangle is 5 units by 3 units on a grid. Each unit is 4 meters. Find its real area.

Real width: 5 × 4 = _____ m.

Real height: 3 × 4 = _____ m.

Answer: Area = 20 × 12 = _____ m²

You try

1. On a plan, 1 unit = 5 ft.
a. 8 units = _____ ft
b. 35 ft = _____ units

_____

2. A rectangle has an area of 600 ft². One side is 20 ft. Find the other side.

_____

Sum it up: To find a missing dimension with a scale, I ___ because ___.

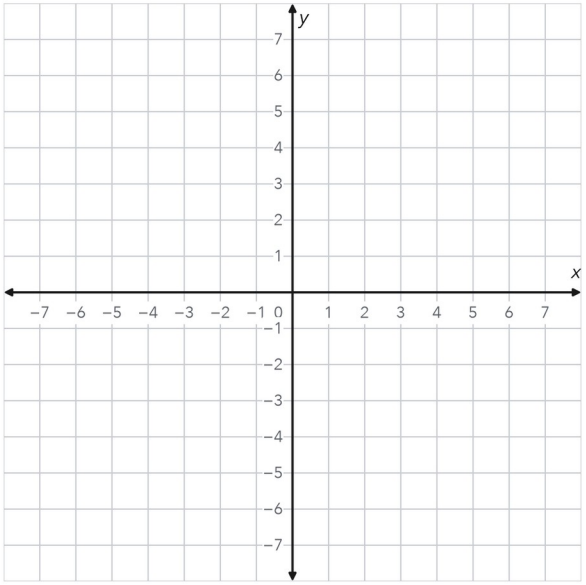
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Session 1 · Model, reason, and explain

Polygons and Perimeter on the Plane · 6.G.A.3

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Draw the rectangle (−2,3),(4,3),(4,−1),(−2,−1). Find perimeter and area, then explain a missing corner if only the first three vertices were supplied.



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Extend your reasoning

Create a contrasting example that exposes the misconception: confuses coordinate values with side lengths.. Explain what changes and what stays true.

_____

_____

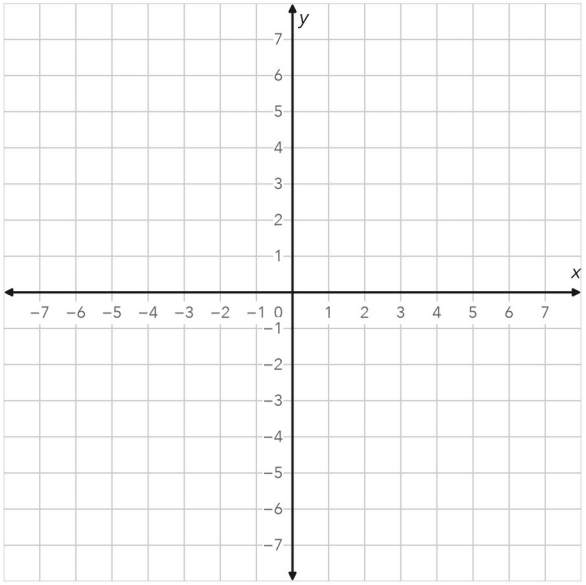
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Designing Polygons with Scale · 6.G.A.3

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A grid rectangle is 5 units by 3 units. Each unit is 4 m. Find real perimeter and area. Explain why area does not simply multiply by 4.



_____

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_____

Extend your reasoning

Create a contrasting example that exposes the misconception: applies a length scale only once to area.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?